

CIS — Nova Bank Online v1.0

Component Integration Specification

Field	Value
Document ID	CIS-NOVA-BANK-ONLINE-v1
Version	1.0
Status	Draft
Platform	OneX Bridge / NSB Online Banking
Customer brand	Nova Bank Online
Internal service	onex-production-platform / onex-ledger-middleware

1. Purpose and scope

This CIS defines how **Nova Bank Online** integrates with the OneX production platform. Nova Bank Online is the customer-facing sovereign online banking brand; the underlying implementation is the OneX **NSB (National Sovereign Bank)** online bank module with IBAN accounts, fiat payment rails, core-banking sync, and HYBX exchange middleware.

In scope

- Online bank accounts, transfers, deposits, statements
- Fund classes M0, M1, NSB
- Payment rails: internal, IBAN, SEPA, SWIFT, wire, ACH, FPS
- Integrations: Fineract core banking, HYBX exchange, virtual cards, Bridge7 ledger, cash codes
- REST API contract on the OneX bridge (:9338)

Out of scope

- Nova 1 Chain settlement (see CIS-Nova-1-Chain-22016-v1.md)
- Z Bank M1–M4 operational liquidity model (see CIS-Z-Bank-Online-v1.md)
- Mobile app store distribution
- Third-party KYC/AML vendor selection

2. System identifiers

Identifier	Value
Bank display name	Nova Bank Online
Legal / ledger alias	NSB — National Sovereign Bank
SWIFT/BIC (default)	NSBKLAL2X
Bank ID (routing)	nsb
Fund class	nsb
Production domain (primary)	zblockchainsystem.com
Trustee / alternate domain	Redirects to zblockchainsystem.com
Bridge listen port	9338
Wallet UI path	/wallet/#onlinebank

Nova Bank Online reuses the NSB ledger namespace. The persisted online-bank state (~/ .onex/online-bank.json) may

override `name` and `swift` for Nova branding at first seed.

3. Architecture

[Architecture diagram — see markdown source]

```
flowchart TB
    subgraph client [Client layer]
        W[Wallet UI /wallet/#onlinebank]
        M[Mobile WebView wallet]
        E[External API consumers]
    end

    subgraph bridge [OneX Bridge :9338]
        OB[Online Bank handlers /bridge/bank/*]
        HY[HYBX middleware /bridge/bank/hybx/*]
        FC[Fineract sync /bridge/bank/finact/*]
        SW[SWIFT release /bridge/bank/swift/*]
        VC[Virtual cards /bridge/cards/*]
        B7[Bridge7 /bridge/bridge7/*]
        LG[Ledger middleware /bridge/ledger/*]
    end

    subgraph data [Persistence]
        OL[online-bank.json]
        BL[bank-ledger.json]
        RL[real ledger book]
    end

    subgraph external [External systems]
        FIN[Fineract core banking]
        HYX[HYBX API]
        PL[Plaid / TrueLayer optional]
    end

    W --> OB
    M --> OB
    E --> OB
    OB --> OL
    OB --> BL
    OB --> HY
    OB --> FC
    HY --> HYX
    FC --> FIN
    LG --> RL
    B7 --> RL
```

4. Environment matrix

Environment	Host	Ledger mode	Online bank
Local dev	127.0.0.1:9338	demo	ONEX_ONLINE_BANK=1
Production	zblockchainsystem.com	production	ONEX_ONLINE_BANK=1
Trustee (legacy host)	redirects → zblockchainsystem.com	production	ONEX_ONLINE_BANK=1

4.1 Required environment variables

```
ONEX_LEDGER_MODE=production
ONEX_ONLINE_BANK=1
ONEX_BANK_LEDGER_FILE=configs/bank-ledger.nova.example.json
ONEX_API_KEY=<long-random-secret>
ONEX_PRODUCTION_DOMAIN=zblockchainsystem.com
ONEX_CORS_ORIGINS=https://zblockchainsystem.com,https://www.zblockchainsystem.com
```

4.2 Optional integrations

```
# HYBX exchange middleware
ONEX_HYBX_ENABLED=1
ONEX_HYBX_URL=https://api.hybrix.io

# Fineract core banking
ONEX_FINERACT_ENABLED=1
ONEX_FINERACT_URL=https://fineract.hybxfinance.com/fineract-provider
ONEX_FINERACT_TENANT=default
ONEX_FINERACT_USERNAME=<username>
ONEX_FINERACT_PASSWORD=<password>

# Bridge7 multi-ledger merge
ONEX_BRIDGE7_ENABLED=1
ONEX_BRIDGE7_PATHS_FILE=configs/bridge7.paths.json

# Cash codes + virtual cards
ONEX_CASHCODE_ENABLED=1

# Open banking providers (optional)
ONEX_BANK_PROVIDER=file
# ONEX_PLAID_CLIENT_ID=...
# ONEX_TRUELAYER_ACCESS_TOKEN=...
```

Full production template: `deploy/env.zblockchainsystem.com.example`

Legacy hosts redirect — see `deploy/CANONICAL-DOMAIN.md`

5. Data models

5.1 Online bank account

Field	Type	Description
<code>id</code>	string	Account ID (e.g. <code>nova-usd-checking</code>)
<code>name</code>	string	Display name
<code>iban</code>	string	IBAN or Nova-format account number
<code>currency</code>	string	ISO 4217 (USD, EUR, GBP, ...)
<code>balance</code>	string	Decimal balance
<code>fundClass</code>	string	<code>m0</code> , <code>m1</code> , <code>nsb</code>
<code>bank</code>	string	<code>nsb</code> , <code>nova</code> , <code>central</code>
<code>status</code>	string	Account status

5.2 Transaction types

type / rail	Description
internal	Between Nova Bank accounts
iban	IBAN transfer
sepa	SEPA credit transfer
swift	SWIFT MT release
wire	Domestic wire
ach	ACH (US)
fps	Faster Payments (UK)

5.3 Fund classes

Class	Label	Use
m0	M0 — base money	Central reserves, vault
m1	M1 — demand deposits	Checking, transactional
nsb	NSB / Nova sovereign	Sovereign reserve accounts

Seed data: `configs/bank-ledger.nova.example.json`

6. API contract

Base URL: `https://<HOST>/bridge` (or `http://127.0.0.1:9338/bridge` locally)

Authentication: `X-API-Key: <ONEX_API_KEY>` on mutating endpoints where enforced.

6.1 Online bank core

Method	Endpoint	Purpose
GET	<code>/bridge/bank/status</code>	Bank health, provider, integrations
GET	<code>/bridge/bank/accounts</code>	List all accounts
GET	<code>/bridge/bank/account?id=</code>	Single account
GET	<code>/bridge/bank/transactions</code>	History (<code>?account=</code> , <code>?type=</code> , <code>?limit=</code>)
GET	<code>/bridge/bank/ledger</code>	Ledger snapshot by fund class
GET	<code>/bridge/bank/wire?account=</code>	Wire instructions
GET	<code>/bridge/bank/statement?account=</code>	CSV statement export
POST	<code>/bridge/bank/transfer</code>	Internal or IBAN transfer
POST	<code>/bridge/bank/send</code>	Alias for transfer
POST	<code>/bridge/bank/deposit</code>	Credit account

6.2 Transfer request body

```
{
  "fromAccount": "nova-usd-checking",
  "toAccount": "nova-eur-transactional",
  "amount": "1000.00",
  "rail": "internal",
  "toIban": "",
  "reference": "Nova transfer ref",
  "preview": false
}
```

6.3 Deposit request body

```
{
  "toAccount": "nova-usd-sovereign",
  "amount": "5000.00",
  "source": "wire",
  "reference": "Incoming wire",
  "preview": false
}
```

6.4 HYBX exchange middleware

Method	Endpoint	Purpose
GET	/bridge/bank/hybx/status	HYBX client health
GET	/bridge/bank/hybx/middleware/status	Middleware status
GET	/bridge/bank/hybx/exchange/routes	Available exchange routes
GET	/bridge/bank/hybx/exchange/quote	Quote for pair
POST	/bridge/bank/hybx/exchange	Execute Nova ↔ HYBX ↔ chain exchange
POST	/bridge/bank/hybx/settle	Settle exchange to destination

6.5 Fineract core banking

Method	Endpoint	Purpose
GET	/bridge/bank/fineract/status	Fineract connection
GET	/bridge/bank/fineract/accounts	Core banking accounts
POST	/bridge/bank/fineract/sync	Sync Fineract → Nova online bank
POST	/bridge/bank/fineract/deposit	Deposit via Fineract
POST	/bridge/bank/fineract/withdraw	Withdraw via Fineract

6.6 SWIFT release

Method	Endpoint	Purpose
GET	/bridge/bank/swift/status	SWIFT system status
POST	/bridge/bank/swift/release	Release SWIFT payment

6.7 Virtual cards

Method	Endpoint	Purpose
GET	/bridge/cards/status	Cards 101.1 status
POST	/bridge/cards/issue	Issue virtual card
POST	/bridge/bank/hybx/cards/issue	Issue via HYBX

6.8 Production status (unified)

Method	Endpoint	Purpose
GET	/bridge/production/status	Full platform status + API map
GET	/bridge/health/green	Green health check

7. Security and connectivity

Control	Requirement
TLS	Required in production (zblockchainsystem.com)
API key	ONEX_API_KEY — rotate on compromise
CORS	Restrict to approved origins via ONEX_CORS_ORIGINS
Bank ledger file	0600 permissions; never commit live balances
Fineract credentials	Store in .env or /etc/onex/onex.env only

8. Deployment

8.1 VPS deploy (production)

```
cd /opt/onex
git pull
cp deploy/env.zblockchainsystem.com.example .env
# Edit ONEX_API_KEY, Fineract creds, bank ledger path
docker compose -f docker-compose.prod.yml up -d --build
```

Canonical domain: `deploy/CANONICAL-DOMAIN.md`

8.2 Verification checklist

```
curl -s https://HOST/bridge/production/status | jq '.onlineBank'
curl -s https://HOST/bridge/bank/status | jq .
curl -s https://HOST/bridge/bank/accounts | jq '.count'
curl -s https://HOST/bridge/bank/ledger | jq '.byFundUsd'
```

Acceptance criteria

- `onlineBank.online = true`
- At least one Nova/NSB account seeded
- `hybxMiddleware.enabled = true` when HYBX configured
- `fineract.enabled = true` when Fineract configured
- Transfer preview returns `status: "ok"` without error

9. Cross-system flows

Flow	Mechanism
Fiat deposit → ledger	POST /bridge/bank/deposit → Fineract sync
Nova → crypto payout	POST /bridge/ledger/settle with externalTo: "nova-1:0x..."
Nova ↔ HYBX mirror	POST /bridge/bank/hybx/exchange route nsb-hybx
Card spend → debit	Virtual card middleware debits online bank account
Multi-ledger merge	POST /bridge/bridge7/sync

See CIS-Nova-Integration-Matrix-v1.md for Nova 1 Chain 22016 settlement paths.

10. Appendices

A. Sample status response (abbreviated)

```
{
  "name": "Nova Bank Online",
  "online": true,
  "swift": "NSBKLAL2X",
  "accounts": 6,
  "enabled": true,
  "provider": { "provider": "file" },
  "hybx": { "enabled": true },
  "fineract": { "enabled": true }
}
```

12. Payment Gateway (card acquiring)

Nova Bank Online includes an integrated **Payment Gateway** for Visa, Mastercard, and American Express via a certified acquirer/processor adapter (Stripe in production, mock in development).

Item	Value
Portal URL	/payments/ (hosted on onex-bridge)
Framework	nova (default) or zbank
Config file	configs/payment-gateway.example.json
Flows	donation, payment, collection
Settlement	Internal Nova Bank accounts or external nominated banks (Wells Fargo, ANZ, Citibank, Lloyds, etc.)

12.1 API routes

Method	Path	Description
GET	/bridge/payments/status	Gateway status
GET	/bridge/payments/config	Public config (framework, fees, Stripe publishable key)
GET	/bridge/payments/pages	Active hosted pages
GET	/bridge/payments/page?slug=	Single page config
POST	/bridge/payments/session	Create payment session
POST	/bridge/payments/confirm	Confirm payment (mock / post-3DS)
POST	/bridge/payments/webhook	Processor webhook (Stripe signature verified)

12.2 Settlement routing

Each payment page references a `settlementDestination` ID. Destinations may be:

- **internal** — credit a Nova Bank Online ledger account immediately
- **external** — credit gateway clearing account; payout to nominated bank via acquiring settlement (subject to processor and regulatory rules)

Optional **processing fees** are configurable globally or per-page (`percent` + `fixed`).

12.3 Environment variables

Variable	Purpose
<code>ONEX_PAYMENT_GATEWAY</code>	Enable gateway (<code>1</code>)
<code>ONEX_PAYMENT_GATEWAY_FILE</code>	Path to gateway config JSON
<code>ONEX_PAYMENT_GATEWAY_FRAMEWORK</code>	<code>nova</code> or <code>zbank</code>
<code>ONEX_PAYMENT_GATEWAY_PROVIDER</code>	<code>mock</code> or <code>stripe</code>
<code>ONEX_STRIPE_SECRET_KEY</code>	Stripe secret key
<code>ONEX_STRIPE_PUBLISHABLE_KEY</code>	Stripe publishable key
<code>ONEX_STRIPE_WEBHOOK_SECRET</code>	Stripe webhook signing secret

B. Related documents

Document	Path
Nova 1 Chain CIS	<code>docs/cis/CIS-Nova-1-Chain-22016-v1.md</code>
Integration matrix	<code>docs/cis/CIS-Nova-Integration-Matrix-v1.md</code>
Bridge7 handoff	<code>deploy/BRIDGE7-SHARE.md</code>
Production env	<code>deploy/env.zblockchainsystem.com.example</code>
Bank ledger seed	<code>configs/bank-ledger.nova.example.json</code>

C. Error handling

HTTP	Meaning
400	Missing required field (<code>id</code> , <code>account</code> , malformed JSON)
405	Wrong HTTP method
200 + <code>error</code>	Business logic failure (insufficient balance, unknown account)